dsDNA
Core
Logo

DSU dsDNA Core Undergraduate Research Training Program

Using uafR for Reproducible Chemical and Mass Spectrometry Research

A 10-week applied research curriculum for undergraduate interns

Program Promise

Students finish this program with a reproducible research project, defensible chemical annotations, interpretable trait matrices, publication-style figures, and a clear plan for communicating results to scientific audiences.

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Preface

This program trains undergraduate interns to move from raw chemical observations to reproducible, interpretable research products. It is built around uafR because the package connects GC-MS processing, public chemical databases, and analysis-ready chemical traits. The training is intentionally broader than one software tool: students also learn project organization, literature interpretation, figure design, peer review, and manuscript-style scientific communication.

This second-pass version is written as a full-day, 10-week research internship. It assumes students are present most weekdays and are expected to read, code, analyze, write, meet with the dsDNA Core team, revise, and produce tangible research artifacts. If the program is run part-time, the deliverables stay intact and the number of exercises is reduced rather than compressing every activity.

Who This Is For

The curriculum is designed for undergraduate interns from chemistry, biology, agriculture, environmental science, public health, data science, and related fields. No student needs to arrive with a finished project. The program helps them bring their own interests into a shared research workflow.

Program Outcomes

By the end of 10 weeks, each student should be able to:

- Define a research question that can be answered with chemical identities, chemical traits, and reproducible analysis.
- Organize an R project with clear file structure, version control, input data, scripts, outputs, and documentation.
- Use uafR to process tentative chemical identities, enrich query chemicals, and inspect database evidence.
- Convert chemical metadata into clean grouping variables and matrices for exploratory analysis.
- Build figures and tables that support a defensible scientific interpretation.
- Explain uncertainty, database provenance, and limits of tentative chemical annotation.
- Deliver a final research package that can support a poster, internal report, manuscript section, or future grant work.

How To Use This Manual

Each week has the same structure so students always know what is expected:

- **Why this matters**: the scientific reason for the week.
- **Concepts**: the vocabulary and logic students must understand.
- **uafR and R skills**: the specific computational skills taught.
- **Guided lab**: a dsDNA Core-supported activity.
- **Independent extension**: a student-owned application.
- **Deliverables**: concrete products that move the research project forward.
- **Quality checklist**: the standard students must meet before moving on.

How the Program Runs

Student work is reviewed weekly with the dsDNA Core team. The goal is not to demand perfect answers at the first pass. The goal is to train students to produce traceable work: every claim connects to data, code, a database field, or a cited source. When a result looks interesting, the review returns to one question: "What evidence supports that, and what could be wrong?"

Core Principle

The student should never treat database text as a conclusion. Database records are evidence. The research conclusion comes from cleaned variables, source coverage checks, comparison across evidence types, and biological or chemical reasoning.

Recommended Weekly Rhythm

Meeting	Purpose
Monday kickoff	Introduce the week, confirm each student's research question, assign readings, and define the deliverable.
Tuesday or Wednesday technical lab	Work through code, database interpretation, and analysis issues with dsDNA Core support.
Thursday writing and evidence clinic	Convert results into claims, captions, methods text, and documented limitations.
Friday review	Inspect deliverables, update the research log, complete peer review, and set next-week priorities.

Expected Final Package

Every student should finish with:

- A reproducible R project folder.
- A cleaned chemical query list and documented inclusion rules.
- uafR enrichment outputs and validation summaries.
- A trait matrix or grouped chemical table suitable for analysis.
- At least three publication-style figures or tables.
- A short manuscript-style report.
- A presentation or poster outline.
- A research log that documents decisions, errors, and interpretation changes.
- A continuation note that allows the next student to extend the project without starting over.

1 Program Architecture for a Full-Day Internship

1.1 Why This Is a 10-Week Full-Day Program

The program is not 10 weeks because students need 10 weeks to learn a few R functions. It is 10 weeks because the expected outcome is a research product with defensible evidence, reusable code, figures, interpretation, and a handoff package. A full-day internship gives students enough time to repeat the cycle that real research requires: learn, try, fail, diagnose, revise, document, and communicate.

Full-Day Assumption

Each week assumes roughly 30 to 35 focused hours. A typical day includes a short technical lesson, guided work, independent project time, dsDNA Core or peer feedback, and written documentation. Students who only attend part-time can still use the program; in that format the number of exercises is reduced rather than reducing the quality standard.

1.2 Daily Work Blocks

Block	Purpose
Morning concept block	Introduce the scientific idea, database concept, coding pattern, or interpretation principle needed that day.
Guided technical lab	Run code, inspect outputs, troubleshoot errors, and learn the expected workflow with a small example.
Independent project block	Apply the day's method to the student's own project or assigned case study.
Evidence and writing block	Convert analysis into a research log entry, methods text, figure caption, or decision memo.
dsDNA Core checkpoint	Review a concrete artifact and decide whether the student can move forward or must revise.

1.3 Product Ladder

The internship is organized as a product ladder. Each week creates an artifact that is needed later.

Week	Artifact	Why it matters
1	Project brief	Prevents a broad interest from becoming an untestable analysis.
2	Reproducible project	Gives the student a research workspace that can be audited and rerun.
3	Database evidence map	Separates identity, source coverage, biological context, and uncertainty.
4	Core GC-MS workflow	Establishes how tentative chemical evidence enters the project.
5	Enrichment result	Creates the main source-backed chemical evidence object.
6	Grouping dictionary	Converts source-backed fields into analysis-ready variables.
7	Figure set	Reveals patterns and missing data before interpretation hardens.
8	Evidence memo	Connects database evidence to peer-reviewed context.
9	Draft report	Forces the methods, results, figures, and limitations into a coherent argument.
10	Final package	Leaves behind work that the dsDNA Core team, a collaborator, or a future student can use.

1.4 Skill Progression

Students move through four levels:

1. **Operator**: can run a provided script and identify the output.
2. **Inspector**: can check whether an output is complete, duplicated, missing, or suspicious.
3. **Analyst**: can transform validated outputs into variables, figures, and tables.
4. **Researcher**: can defend interpretation with source evidence, literature, uncertainty, and next steps.

1.5 Research Integrity Rules

Rules Students Must Follow

- Never present a tentative chemical match as confirmed identity unless standards or equivalent confirmatory evidence are available.
- Never hide unmatched chemicals or missing source coverage.
- Never copy a database sentence into a result and treat it as an analyzed variable.
- Never store tokens, passwords, private data, or unpublished collaborator data in a repository.
- Never change raw data manually; all cleaning must be scripted or documented.
- Never make a figure that cannot be regenerated from code.

1.6 Program Decision Gates

Decision gates keep the program rigorous:

Gate	Student must show
Scope gate	A feasible question, reviewed chemical list, and documented risk note.
Reproducibility gate	A project folder with relative paths, scripts, logs, and no secrets.
Evidence gate	Source coverage, validation results, and evidence rows for important traits.
Variable gate	A grouping dictionary with source table, field, rule, allowed values, and caveat.
Interpretation gate	A claim supported by a figure, table, database field, and literature source.
Handoff gate	Final package can be understood by someone who was not present during the internship.

1.7 Expected Weekly Time Budget

Activity	Hours/week	Notes
Technical instruction and guided lab	6–8	Short lessons followed by hands-on work.
Independent project work	10–12	Student applies the workflow to their question or case study.
Reading and annotation	3–5	Database docs, methods papers, and project-specific literature.
Writing and documentation	4–6	Research log, methods, captions, interpretation, and handoff notes.
dsDNA Core and peer review	3–4	Weekly review, code walk-throughs, and figure critique.
Troubleshooting and revision	4–6	Expected time for errors, reruns, cleanup, and rethinking.

1.8 What Students Should Be Told Up Front

The internship is not a race to get a colorful heatmap. A student who finds that half their chemicals have weak source coverage has not failed. That is a useful result if it is documented clearly. A student who discovers that a project question is too broad has not failed. That is research judgment. The standard is not that every project produces a dramatic biological conclusion. The standard is that every project produces a trustworthy research package.

2 Full-Day 10-Week Schedule

This chapter provides the day-by-day structure that turns the program into a full-day internship. The order can be adjusted for local scheduling, but the sequence remains intact: scope before analysis, validation before interpretation, and evidence before claims.

2.1 Week 1: Research Onboarding and Question Design

Day	Morning	Lab or project block	End-of-day product
1	Orientation to DSU dsDNA Core research expectations, safety, data integrity, and responsible conduct.	Students inventory their interests and match them to possible chemical evidence types.	Draft interest statement and initial discussion questions.
2	What makes a chemical research question testable.	Convert broad topics into question, system, comparison, evidence, and limitation fields.	Three candidate research questions.
3	Chemical identity, tentative annotation, and why uncertainty matters.	Build first chemical intake table from literature, standards, GC-MS hits, or program-provided examples.	Chemical intake table v1.
4	Evidence types across PubChem, KEGG, LOTUS, FEMA, FDA/SPL, MeSH, and literature.	Map which evidence types could answer each candidate question.	Evidence feasibility matrix.
5	Scope review and project selection.	Revise question, write risk note, and define Week 2 setup needs.	Reviewed one-page project brief.

2.2 Week 2: R, RStudio, Git, and Reproducible Structure

Day	Morning	Lab or project block	End-of-day product
1	RStudio projects, file paths, and why raw data are protected.	Create project folder and run installation checks.	Project skeleton and install log.
2	Reading and writing tables in R.	Practice importing CSV files, checking dimensions, and writing processed outputs.	Data import script with comments explaining decisions.
3	Git concepts: commits, branches, ignored files, and secrets.	Make first local commits and inspect what should not be committed.	Clean repository status and research log entry.
4	Reproducible scripts and numbered workflows.	Create script sequence for the student's project.	Numbered script plan and README update.
5	Troubleshooting clinic.	Each student reruns setup from a clean R session and documents failures.	Reproducibility checkpoint signed off in program review.

2.3 Week 3: Chemical Data Literacy and Public Databases

Day	Morning	Lab or project block	End-of-day product
1	Names, CIDs, InChIKeys, synonyms, salts, mixtures, stereochemistry, and ambiguity.	Students audit five chemicals for naming risks.	Identifier ambiguity note.
2	PubChem structure: properties, annotations, assays, classifications, and source-specific records.	Explore one compound manually and compare to uafR profile tables.	PubChem evidence worksheet.
3	KEGG biochemical context: compounds, reactions, enzymes, modules, pathways, and academic-use etiquette.	Inspect a KEGG compound record and identify pathway or reaction fields.	KEGG evidence worksheet.
4	Natural product occurrence, biomedical terms, regulatory records, and flavor/sensory context.	Compare LOTUS, MeSH, FEMA, and FDA/SPL evidence roles.	Source comparison table.
5	Database evidence discussion.	Students revise project scope based on source coverage expectations.	Database evidence map v1.

2.4 Week 4: uafR Core GC-MS Workflow

Day	Morning	Lab or project block	End-of-day product
1	GC-MS hit tables, required columns, match factors, retention time, base peaks, and peak area.	Audit example input columns and suspicious values.	Input column audit.
2	Reshaping and extracting chemical hits with uafR.	Run the offline core workflow and inspect bundled outputs.	Annotated core workflow script.
3	Duplicates, ambiguous peaks, and tentative identity limits.	Students identify duplicate or conflicting hits in an example table.	Duplicate and uncertainty note.
4	Applying the workflow to student data or a case dataset.	Run a small project-specific workflow.	Processed output v1.
5	Methods writing clinic.	Write a methods paragraph for GC-MS processing and tentative identification caveats.	Methods paragraph draft.

2.5 Week 5: Research-Grade categorate Enrichment

Day	Morning	Lab or project block	End-of-day product
1	What `categorate(detail = "research")` returns and why validation comes first.	Run a two-compound smoke test when services are available or inspect saved examples.	Validation summary and table inventory.
2	Source coverage and provenance.	Inspect `SourceCoverage`, `SourceDiagnostics`, `TableQuality`, and `DataDictionary`.	Source coverage report.
3	PubChem enrichment outputs.	Trace properties, safety, literature, classifications, and source annotations for selected chemicals.	PubChem evidence audit.
4	KEGG and source-specific enrichment outputs.	Trace pathway, reaction, enzyme, occurrence, and taxonomy rows.	KEGG/LOTUS evidence audit.
5	Scaling and caching.	Run project list in batches, save RDS results, and document query settings.	Saved enrichment object and query log.

2.6 Week 6: Research Variables and Trait Matrices

Day	Morning	Lab or project block	End-of-day product
1	Discrete variables versus descriptive text.	Inspect `ChemicalTraits` and separate analysis-ready values from narrative fields.	Candidate variable list.
2	Confidence, evidence, and matrix keys.	Build binary and confidence-weighted matrices.	Trait matrix v1.
3	Project grouping dictionary.	Define allowed values, source fields, extraction rules, and caveats.	Grouping dictionary v1.
4	Real-world grouping exercise.	Compare two chemical sets by shared and distinct traits.	Comparison table and interpretation note.
5	Variable gate.	Remove weak variables, revise rules, and freeze analysis inputs for Week 7.	Reviewed trait matrix and dictionary.

2.7 Week 7: Visualization and Exploratory Analysis

Day	Morning	Lab or project block	End-of-day product
1	Figure purpose, audience, and visual honesty.	Sketch three figure questions before plotting.	Figure plan worksheet.
2	Coverage and trait heatmaps.	Generate heatmaps and inspect missingness.	Coverage and trait figures v1.
3	Property and trait summaries.	Build bar charts, scatterplots, and summary tables.	Two additional figures or tables.
4	Figure critique.	Peer review labels, captions, color choices, and claim strength.	Revised figure set.
5	Exploratory interpretation.	Write a results paragraph for each figure.	Figure captions and results draft.

2.8 Week 8: Literature Integration and Evidence-Based Interpretation

Day	Morning	Lab or project block	End-of-day product
1	Evidence ladder from tentative match to confirmed conclusion.	Rank the student's major findings by evidence strength.	Evidence ladder table.
2	Literature search strategies.	Build search strings from chemical names, traits, pathways, organisms, and methods.	Search log.
3	Reading scientific papers efficiently.	Extract system, method, chemical evidence, and limitation from three papers.	Annotated literature table.
4	Connecting literature to figures.	Write claim-evidence-limitation paragraphs.	Interpretation memo v1.
5	Interpretation gate.	Remove unsupported claims and identify follow-up experiments.	Reviewed interpretation memo.

2.9 Week 9: Manuscript-Style Research Product

Day	Morning	Lab or project block	End-of-day product
1	Manuscript structure and argument flow.	Build report outline from figures and evidence memo.	Report outline.
2	Methods detail and reproducibility.	Write methods from scripts, settings, package versions, and validation outputs.	Methods section v1.
3	Results without overclaiming.	Write results paragraphs tied to figures and tables.	Results section v1.
4	Discussion, limitations, and next steps.	Draft discussion and follow-up plan.	Discussion section v1.
5	Full draft review.	Peer and dsDNA Core review for logic, reproducibility, and claim strength.	Complete draft report.

2.10 Week 10: Presentation and Final Handoff

Day	Morning	Lab or project block	End-of-day product
1	Scientific presentation structure.	Build slide or poster outline around one central result.	Presentation outline.

Day	Morning	Lab or project block	End-of-day product
2	Visual polishing and narrative flow.	Revise figures, captions, and talking points.	Presentation draft.
3	Reproducibility and handoff audit.	Run final project check and clean file organization.	Handoff package v1.
4	Practice presentation and peer review.	Deliver practice talk, answer questions, revise weak points.	Revised final presentation.
5	Final symposium and archive.	Present, submit final report, and write continuation note.	Complete final research package.

3 Research Onboarding and Question Design

Week 1: Research Onboarding and Question Design

Research milestone: Turn a broad interest into a testable chemical research question.
Primary product: One-page project brief with question, system, candidate chemicals, expected evidence, and risks.

3.1 Why This Matters

Strong computational research starts before any code is run. Students need a clear scientific question, a defensible reason for studying a chemical set, and a plan for evaluating evidence. Without this foundation, database enrichment becomes a list of facts rather than a research workflow.

3.2 Concepts

- **Research system**: the organism, environment, treatment, material, food, exposure, sample type, or process being studied.
- **Chemical evidence**: tentative identifications, mass spectral matches, database annotations, literature support, functional groups, biological activities, hazard terms, occurrence records, pathways, reactions, and measured properties.
- **Analysis-ready variable**: a clean field that can be grouped, counted, filtered, modeled, or visualized.
- **Provenance**: the source and retrieval context for a piece of evidence.
- **Uncertainty**: the gap between a database hit and a confirmed scientific conclusion.

3.3 uafR and R Skills

Students do not need to run the full workflow in Week 1. They should learn what the tool can produce:

- Core GC-MS workflow: `spreadOut()`, `mzExacto()`, and `exactoThese()`.

- Chemical enrichment: `categorate(detail = "research")`.
- Research trait tables: `ChemicalTraits`, `ChemicalTraitOntology`, and `ChemicalTraitMatrix`.
- Hazard summary table: `ChemicalHazards`.
- Occurrence summary table: `ChemicalOccurrences`.
- Measurement summary table: `ChemicalMeasurementSummary`.
- KEGG-specific interpretation table: `KEGGReactionParticipants`.
- Quality checks: `validateCategorateResult()` and `TableQuality`.

3.4 Guided Lab

1. Choose one research domain: plant metabolites, environmental exposure, food chemistry, microbial products, pharmacology, materials, forensic chemistry, toxicology, or another reviewed topic.
2. Write three possible research questions. Each question must name the system, chemicals or chemical class, comparison, and expected evidence.
3. Identify a starting chemical list. This can come from GC-MS results, literature, standards, known metabolites, or a small exploratory list.
4. Mark each chemical as known, tentative, uncertain, or exploratory.
5. Write a risk note: what could make the analysis misleading?

Example Research Questions

- Do volatile compounds detected in two plant treatments differ in sensory trait profiles, natural product occurrence, or pathway roles?
- Which detected compounds have safety or exposure signals that require additional review before biological interpretation?
- Do compounds associated with a sample group cluster by functional group, pathway, or literature-backed biological activity?

3.5 Independent Extension

Students interview the program lead, the dsDNA Core mentor, or a project lead for 20 minutes and capture:

- The research context.
- Why the chemical list matters.
- What decisions the analysis should support.
- What evidence would be convincing.
- What evidence would be weak or unacceptable.

3.6 Deliverables

Deliverable	Evidence of quality	Submission format
Project brief	Includes question, system, comparison, candidate chemical list, and expected output.	PDF or Markdown
Chemical intake table	Has one row per chemical and columns for source, confidence, project relevance, and notes.	CSV
Risk note	Names at least three ways the analysis could be wrong or overinterpreted.	Research log

3.7 Quality Checklist

Before Week 2

- The question is narrow enough to analyze in 10 weeks.
- The chemical list has a clear source and inclusion rule.
- The student can explain why chemical traits, database evidence, or GC-MS patterns matter for the question.
- The project scope has passed dsDNA Core review.

4 R, RStudio, Git, and Reproducible Project Structure

Week 2: R, RStudio, Git, and Reproducible Project Structure

Research milestone: Build a project that another researcher can open, run, and audit.
Primary product: Working R project with folders, install check, research log, and first commit.

4.1 Why This Matters

Reproducibility is a research skill, not an afterthought. A student who can organize files, name outputs clearly, and rerun an analysis has already solved many of the problems that slow down publication.

4.2 Concepts

- **Project root**: the top-level folder that contains scripts, data, results, and documentation.
- **Relative path**: a file path that works from the project root on any machine.
- **Raw data**: input data that should not be manually edited.
- **Derived data**: tables created by scripts.
- **Version control**: a record of changes to files and analysis decisions.
- **Research log**: a dated record of decisions, errors, fixes, and interpretation changes.

4.3 Recommended Project Structure

```
student-project/
  README.md
  analysis/
  data_raw/
  data_processed/
  figures/
  results/
```

```
scripts/
reports/
logs/
```

4.4 Guided Lab

Run the install and environment check in `training/scripts/00_install_check.R`. Students should save the output in their research log.

```r
source("training/scripts/00_install_check.R")
check_training_install()
```

Then create a project folder and initialize the file structure:

```r
source("training/scripts/01_project_setup.R")
create_training_project("student-project")
```

4.5 Git Practice

Students should make small, readable commits. A good commit message says what changed and why it matters:

```
Add chemical intake table for plant volatile project
Document first uafR install check
Add Week 1 project brief
```

4.6 Independent Extension

Students customize their project README with:

- Project title.
- Research question.
- Data source.
- Main scripts.
- Contact and dsDNA Core support.
- Current status.

4.7 Deliverables

Deliverable	Evidence of quality	Submission format
Project folder	Uses the required folder structure and relative paths.	Local folder or repository
Install check	Shows R version, uafR version, key dependencies, and package data availability.	Research log
First commit	Commit includes Week 1 and Week 2 materials without private tokens or sensitive data.	Git history

4.8 Quality Checklist

Before Week 3

- No script uses an absolute path to another person's computer.
- No token, password, or private key appears in any file.
- The project can be opened from the project root and scripts can find their inputs.
- The student can explain the difference between raw and processed data.

5 Chemical Data Literacy and Public Database Foundations

Week 3: Chemical Data Literacy and Public Database Foundations

Research milestone: Learn what public chemical databases can and cannot prove.
Primary product: Database evidence map for the student's chemical list.

5.1 Why This Matters

Public databases are powerful, but they are not interchangeable. PubChem, KEGG, LOTUS, FEMA, FDA/SPL, MeSH, and literature records answer different questions. Students need to know which source supports which type of claim.

5.2 Database Roles

Source	Useful evidence	Important caution
PubChem	Identifiers, synonyms, properties, annotations, hazards, classifications, assays, literature links.	Records can aggregate broad text from many sources; verify source-specific context.
KEGG	Compound IDs, reactions, pathways, enzymes, modules, biochemical context.	Coverage is strongest for curated biochemical systems; absence is not evidence of no role.
LOTUS	Natural product occurrence and taxonomic context.	Occurrence records can be sparse or uneven across organisms.
FEMA	Flavor-use context and sensory relevance.	Flavor presence does not prove abundance or biological role in a sample.
FDA/SPL	Drug product, route, dosage form, and regulatory text.	Regulatory annotation may describe a formulation context rather than the chemical alone.
MeSH	Pharmacologic actions and biomedical terms.	Terms may reflect literature indexing rather than direct mechanism.

5.3 Concepts

- **Identifier ambiguity**: the same name can map to multiple structures or salts.
- **Synonym noise**: synonyms include trade names, old names, registry names, abbreviations, and mixture names.
- **Annotation source**: a database field should be interpreted through its source and section.
- **Ground-truth extraction**: clean values must come from traceable source fields, not guesses.
- **Negative space**: missing data can mean no record, no query match, no source coverage, or a real absence.

5.4 Guided Lab

Students inspect three chemicals from different contexts. For each chemical, they answer:

- What are the stable identifiers?
- What synonyms might confuse a search?
- What database evidence is relevant to the project?
- Which fields are clean enough for grouping?
- Which fields are only useful as narrative evidence?

```
library(uafR)
data("library_data", package = "uafR")

compounds <- c("aspirin", "caffeine", "methyl salicylate")

profile <- pubchemProfile(
  compounds = compounds,
  profile = "minimal",
  cache = TRUE,
  throttle = 0.1
)

profile$identity
profile$properties
head(profile$synonyms)
```

5.5 Independent Extension

Students create a source evidence map:

- Rows are chemicals.
- Columns are evidence types: identity, properties, hazard, occurrence, pathway, reaction, sensory, biomedical, literature, assay, and regulatory.
- Cells contain yes, no, uncertain, or not relevant.

5.6 Deliverables

Deliverable	Evidence of quality	Submission format
Evidence map	Shows source coverage by chemical and evidence type.	CSV or spreadsheet
Ambiguity note	Identifies naming or identifier problems for at least three chemicals.	Research log
Database interpretation paragraph	Explains which sources are most relevant to the project and why.	Report draft

5.7 Quality Checklist

Before Week 4

- The student can explain why a PubChem annotation, KEGG pathway, and LOTUS occurrence do not mean the same thing.
- The chemical intake table has stable names or identifiers whenever available.
- The project has a plan for handling unmatched or ambiguous chemicals.

6 uafR Core GC-MS Workflow

Week 4: uafR Core GC-MS Workflow

Research milestone: Convert GC-MS-style hit tables into a form that supports targeted extraction.
Primary product: Processed example workflow using `spreadOut()`, `mzExacto()`, and `exactoThese()`.

6.1 Why This Matters

Before enrichment, students need to understand the chemical identity table coming from a mass spectrometry workflow. Tentative compound names, retention time, base peak, match factor, file name, and peak area must be reshaped carefully so chemical evidence can be compared across samples.

6.2 Required Input Columns

The core workflow expects a GC-MS-style input table with these columns:

- `Component.RT`
- `Component.Area`
- `Base.Peak.MZ`
- `File.Name`
- `Compound.Name`
- `Match.Factor`

6.3 Guided Lab

Use the package example data first. Students should not begin with their own messy file until they can run and explain the example. The support script uses bundled exact-match output by default so the classroom workflow does not depend on live services.

```r
library(uafR)
source("training/scripts/04_core_workflow.R")
```

```r
core <- run_core_workflow(live_lookup = FALSE)

names(core$spread)
head(core$exact)
```

When services are available and live lookup behavior needs to be tested, run:

```r
live_core <- run_core_workflow(live_lookup = TRUE)
head(live_core$exact)
```

Then subset an existing categorated object:

```r
data("standard_categorated", package = "uafR")
subset_reactives <- exactoThese(
  standard_categorated,
  subsetBy = "Database",
  subsetArgs = "reactives"
)
subset_reactives
```

6.4 Interpretation Practice

Students should describe:

- Which compounds were retained and why.
- Whether match factors are high enough for the intended claim.
- Whether retention time or base peak patterns suggest duplicated or ambiguous hits.
- What should be considered tentative until confirmed by standards.

6.5 Independent Extension

Students adapt the workflow to their own data or a program-provided example. They must document every cleaning step and preserve the original input file unchanged.

6.6 Deliverables

Deliverable	Evidence of quality	Submission format
Core workflow script	Runs from project root and produces spread and exact outputs.	R script
Input column audit	Confirms required columns, missing values, duplicates, and unusual match factors.	CSV or log
Interpretation note	Explains what can and cannot be concluded from the processed hits.	Research log

6.7 Quality Checklist

Before Week 5

- The script preserves raw data and writes processed outputs separately.
- The student can explain why tentative library matches are not structural confirmation.
- Duplicate or suspicious hit keys are documented.
- The workflow uses clear object names and comments only where they clarify decisions.

7 Research-Grade categorate Enrichment

Week 5: Research-Grade categorate Enrichment

Research milestone: Use public database enrichment to convert chemical names into traceable research tables.
Primary product: Validated `categorate(detail = "research")` result and source coverage report.

7.1 Why This Matters

The upgraded `categorate()` workflow gives students much more than a simple database lookup. It returns source-specific tables, normalized chemical traits, ontology-style terms, measurement summaries, hazard fields, occurrence evidence, KEGG roles, quality diagnostics, and data dictionaries. This is where the chemical list becomes a research dataset.

7.2 Key Output Families

- **Identity and properties**: `PubChemIdentity`, `PubChemProperties`, `PubChemIdentifiers`.
- **Source profiles**: `SafetyProfile`, `FEMAProfile`, `FDA_SPL_Profile`, `LOTUSProfile`, `MeSHProfile`, `LiteratureProfile`.
- **Discrete trait data**: `ChemicalTerms`, `ChemicalTraits`, `ChemicalTraitOntology`.
- **Analysis matrices**: `ChemicalTraitMatrix`, `ChemicalTraitOntologyMatrix`.
- **Research summaries**: `ChemicalTraitReport`, `ChemicalMeasurementSummary`, `SourceCoverage`, `DerivedGroups`.
- **Database quality**: `DataDictionary`, `TableQuality`, `SourceDiagnostics`, `ValidationSummary`.

7.3 Guided Lab

Run a small enrichment first. Use caching for routine work and set a gentle throttle when live services are queried.

```r
library(uafR)
data("library_data", package = "uafR")

compounds <- c("aspirin", "caffeine", "methyl salicylate")

result <- categorate(
  compounds = compounds,
  chemical_library = library_data,
  input_format = "wide",
  detail = "research",
  cache = TRUE,
  throttle = 0.2,
  assay_detail_limit = 0,
  trait_matrix_profile = "core",
  trait_matrix_min_confidence = "medium",
  trait_matrix_max_traits = 100
)

validateCategorateResult(result)$Summary
result$SourceCoverage
result$ChemicalTraitReport
result$ChemicalTraitMatrix
```

7.4 Evidence Inspection

Students should not trust a matrix until they inspect the supporting evidence.

```r
trait_key <- names(result$ChemicalTraitMatrix)[2]

evidence <- chemicalTraitEvidence(result)
evidence[evidence$MatrixKey == trait_key, ]
```

7.5 Independent Extension

Students run `categorate()` on their reviewed chemical list. If the list is long, they should begin with 5 to 10 representative chemicals, inspect output quality, then scale up.

7.6 Deliverables

Deliverable	Evidence of quality	Submission format
Enrichment object	Saved result contains validation tables, source coverage, and trait matrices.	RDS
Coverage report	Summarizes which sources returned useful data by chemical.	CSV or table
Evidence audit	Traces five matrix traits back to source evidence.	Research log

7.7 Quality Checklist

Before Week 6

- `ValidationSummary` has been inspected.
- Missing source data are documented rather than silently ignored.
- At least five high-value traits have been traced to evidence rows.
- The student can identify which output tables are raw-ish source evidence and which are normalized analysis tables.

8 Turning Chemical Metadata Into Research Variables

Week 6: Turning Chemical Metadata Into Research Variables

Research milestone: Convert enrichment outputs into clean variables for grouping, filtering, and comparison.
Primary product: Analysis-ready trait table and project-specific grouping dictionary.

8.1 Why This Matters

Full sentences are difficult to analyze. Researchers need discrete values: hazard codes, pathway IDs, organism names, property bins, pharmacologic actions, measurement units, reaction roles, and presence or absence of curated traits. Week 6 teaches students to build those variables without inventing unsupported labels.

8.2 Concepts

- **Trait type**: a broad evidence domain such as safety, sensory, pathway, taxonomy, bioactivity, or physicochemical behavior.
- **Trait value**: the specific clean value, such as a GHS code, KEGG pathway, organism, or functional term.
- **Matrix key**: a stable column name used in a wide analysis matrix.
- **Confidence**: a source-aware estimate of how reliable or direct an extracted trait is.
- **Project grouping dictionary**: the student's reviewed mapping from traits to analysis groups.

8.3 Guided Lab

Start with the normalized tables:

```
traits <- result$ChemicalTraits
ontology <- result$ChemicalTraitOntology
matrix <- result$ChemicalTraitMatrix
```

```
table(traits$TraitType, useNA = "ifany")
table(ontology$OntologyDomain, useNA = "ifany")

head(traits[order(traits$Query, traits$TraitType), ])
```

Build a focused matrix:

```
core_matrix <- chemicalTraitMatrix(
  result,
  profile = "core",
  mode = "binary",
  min_confidence = "medium",
  max_traits = 100
)

confidence_matrix <- chemicalTraitMatrix(
  result,
  profile = "full",
  mode = "confidence",
  min_confidence = "medium",
  max_traits = 200
)
```

8.4 Grouping Rules

Every grouping variable must pass four tests:

1. It is useful for the research question.
2. It is extracted from a traceable source field.
3. It has a reproducible rule.
4. It does not overstate what the database evidence proves.

Good Versus Weak Grouping Variables

Good: A binary variable marking chemicals with source-backed GHS hazard codes. The evidence row stores the code and source.
Weak: A manually assigned variable called "dangerous" based on reading broad safety sentences.

8.5 Independent Extension

Students design a project-specific grouping dictionary with 8 to 20 variables. Each row should include:

- Variable name.
- Source table.
- Source field.

- Extraction rule.
- Allowed values.
- Confidence rule.
- Reason the variable matters.

8.6 Deliverables

Deliverable	Evidence of quality	Submission format
Trait table	Cleaned long table with selected traits and confidence fields.	CSV
Trait matrix	Wide matrix ready for visualization or modeling.	CSV
Grouping dictionary	Documents how each project variable was derived.	CSV or spreadsheet

8.7 Quality Checklist

Before Week 7

- No grouping variable is based only on unsupported intuition.
- The student can trace each selected variable to a source table and evidence row.
- Matrix columns are named clearly enough to interpret later.
- Missing values are handled deliberately.

9 Visualization and Exploratory Analysis

Week 7: Visualization and Exploratory Analysis

Research milestone: Use figures to reveal patterns, gaps, and hypotheses without overclaiming.
Primary product: At least three publication-style exploratory figures with captions.

9.1 Why This Matters

Figures are where students often discover whether the research question is working. A good exploratory figure makes patterns visible, but also shows uncertainty, missing data, and source coverage. Week 7 trains students to make figures that can survive dsDNA Core review.

9.2 Figure Types

Figure	Use
Source coverage heatmap	Shows which databases contributed evidence for each chemical.
Trait matrix heatmap	Groups chemicals by shared discrete traits.
Property scatterplot	Compares measured or computed properties such as exact mass, molecular weight, XLogP, or TPSA when available.
Bar chart	Summarizes trait counts, hazard code counts, source hits, or pathway roles.
Network sketch	Shows chemical-trait or chemical-pathway relationships for a small set.

9.3 Guided Lab

Use the script in `training/scripts/07_visualization.R`. It creates example figures from a saved `categorate()` result.

```
source("training/scripts/07_visualization.R")

plots <- build_training_figures(result, output_dir = "figures")
```

```r
names(plots)
```

9.4 Caption Standard

Each figure caption should include:

- What the figure shows.
- Which table or matrix was used.
- What counts as evidence.
- What limitation matters most.
- One interpretation that can be defended.

9.5 Independent Extension

Students produce a figure set that answers one of these:

- Which chemicals share the most traits?
- Which source contributed the most useful evidence?
- Which chemical classes or traits separate sample groups?
- Which compounds need additional confirmation before interpretation?
- Which chemical traits connect most directly to the project question?

9.6 Deliverables

Deliverable	Evidence of quality	Submission format
Figure set	At least three figures, each with a caption and source table.	PNG or PDF
Figure script	Recreates all figures from processed data.	R script
Exploratory interpretation	Names patterns, caveats, and next analysis decisions.	Report draft

9.7 Quality Checklist

Before Week 8

- Every figure can be regenerated from a script.
- Axes, labels, colors, and legends are readable.
- Captions do not claim more than the data support.
- Missing source coverage is visible or discussed.

10 Literature Integration and Evidence-Based Interpretation

Week 8: Literature Integration and Evidence-Based Interpretation

Research milestone: Connect database evidence to peer-reviewed context.
Primary product: Annotated evidence table and interpretation memo.

10.1 Why This Matters

Database enrichment helps students find leads, but publication-quality interpretation requires literature context. A student must distinguish between a chemical being listed in a database, a chemical being measured in a system, and a chemical having a demonstrated role in a biological or environmental process.

10.2 Evidence Ladder

Evidence level	Interpretation strength
Tentative GC-MS match	Suggests possible presence; requires caution and preferably standards.
Database identifier or property	Supports identity, descriptors, and grouping.
Source annotation	Provides context from curated or aggregated source text.
Pathway, reaction, or occurrence evidence	Suggests biochemical, ecological, or taxonomic context.
Peer-reviewed measurement in similar system	Stronger support for relevance to the student's project.
Confirmed standard or targeted validation	Strongest support for chemical presence in the specific sample.

10.3 Guided Lab

Students select three high-priority findings from their Week 7 figures. For each finding, they fill an evidence table:

- Chemical.
- Trait or pattern.
- uafR table and field.
- Source database.
- Literature citation.
- Interpretation.
- Limitation.
- Follow-up test.

10.4 Finding Literature

Use database records to find search terms, not conclusions. Good searches combine:

- Chemical name or CID.
- System, organism, tissue, material, food, environment, disease, or exposure.
- Trait term, pathway, reaction, hazard code, sensory term, or measured property.
- Analytical method such as GC-MS, LC-MS, headspace, metabolomics, or targeted assay.

10.5 Independent Extension

Students write a two-page interpretation memo with:

- One central result.
- Two supporting figures or tables.
- Three literature sources.
- One paragraph on uncertainty.
- One proposed follow-up experiment or analysis.

10.6 Deliverables

Deliverable	Evidence of quality	Submission format
Annotated evidence table	Links database fields, literature, interpretation, and limits.	CSV or spreadsheet
Interpretation memo	Explains one result with evidence and caveats.	PDF
Reference library	Contains cleaned citations for final report.	BibTeX, Zotero, or plain bibliography

10.7 Quality Checklist

Before Week 9

- Claims are tied to a table, figure, or citation.
- The memo distinguishes tentative identity from confirmed presence.
- Literature sources are relevant to the system, not just the compound.
- Uncertainty is written clearly instead of hidden.

11 Manuscript-Style Research Product

Week 9: Manuscript-Style Research Product

Research milestone: Convert the analysis into a concise scientific report.
Primary product: Draft report with abstract, methods, results, figures, and limitations.

11.1 Why This Matters

Writing is part of analysis. A report exposes weak logic, missing methods, unclear figures, and overclaims. Week 9 turns the student's project into a product that can become a poster, internal report, thesis chapter, manuscript section, or grant preliminary result.

11.2 Report Structure

- **Title**: specific, not inflated.
- **Abstract**: question, approach, main result, limitation, and significance.
- **Introduction**: system background and why chemical traits matter.
- **Methods**: data source, uafR workflow, enrichment settings, filtering, visualization, and literature review.
- **Results**: figures and tables with restrained interpretation.
- **Discussion**: what the results suggest, what remains uncertain, and what should be done next.
- **Reproducibility statement**: how to rerun the analysis and where outputs are stored.

11.3 Methods Details To Include

- uafR version.
- R version and operating system.
- Chemical query list source.
- `categorate()` detail level, cache setting, throttle setting, and assay detail limit.

- Trait matrix profile, confidence threshold, and maximum traits.
- Any manual filtering or grouping rules.
- How missing data and unmatched chemicals were handled.

11.4 Guided Lab

Run the reproducibility check:

```
source("training/scripts/09_reproducibility_check.R")

check_training_project(".")
```

Then draft the methods paragraph from script settings and validation summaries.

11.5 Independent Extension

Students complete a 4 to 6 page report. The goal is not length. The goal is a defensible chain from question to data to evidence to interpretation.

11.6 Deliverables

Deliverable	Evidence of quality	Submission format
Draft report	Includes all required sections and at least three figures or tables.	PDF or Word export
Reproducibility check	Confirms project files, scripts, results, and session information.	Log file
Peer review notes	Student gives and receives structured feedback.	Worksheet

11.7 Quality Checklist

Before Week 10

- The methods section is specific enough to rerun the analysis.
- Results are separated from speculation.
- Each figure is cited in the text.
- The discussion includes limitations and next steps.

12 Presentation, Peer Review, and Final Research Package

Week 10: Presentation, Peer Review, and Final Research Package

Research milestone: Communicate the project clearly and leave behind a reusable research package.
Primary product: Final report, presentation, clean project folder, and continuation note.

12.1 Why This Matters

The final week is about making the work useful to others. A strong undergraduate project should be understandable to a future student, the dsDNA Core team, a collaborator, or a reviewer. The final package should preserve what was done, why it was done, and how to continue.

12.2 Presentation Structure

1. Research question and system.
2. Why the chemical evidence matters.
3. Data source and uafR workflow.
4. Main figure or result.
5. Evidence supporting interpretation.
6. Limitations.
7. Next experiment or analysis.

12.3 Peer Review

Students review each other's final products using the rubric appendix. Reviewers should focus on:

- Clarity of research question.
- Traceability of chemical evidence.

- Quality of cleaned variables and matrices.
- Figure readability.
- Strength of interpretation.
- Reproducibility.

12.4 Final Package Checklist

Required Handoff Contents

- `README.md` explains the project and how to rerun the analysis.
- `data_raw/` contains unchanged input files or instructions for secure access.
- `data_processed/` contains derived tables.
- `scripts/` contains numbered scripts.
- `figures/` contains final figures.
- `results/` contains saved enrichment objects and summary tables.
- `reports/` contains final report and presentation.
- `logs/` contains research log and session information.

12.5 Independent Extension

Students prepare a continuation note:

- What worked.
- What failed.
- Which chemicals or traits deserve follow-up.
- Which database fields were most useful.
- Which results should not be overinterpreted.
- What the next student should do first.

12.6 Deliverables

Deliverable	Evidence of quality	Submission format
Final report	Revised after dsDNA Core and peer feedback.	PDF
Presentation	8 to 12 minute talk or poster-style slide deck.	PDF or slides
Research package	Clean folder with scripts, data, figures, results, and logs.	Project folder
Handoff note	Explains next steps for the project.	Markdown or PDF

12.7 Quality Checklist

Program Completion Standard

- The dsDNA Core team can rerun or inspect the analysis without guessing file locations.
- The final interpretation is evidence-based and appropriately cautious.
- The student can answer where each major result came from.
- The project creates a foundation for continued research.

13 Guided Lab and Exercise Compendium

The weekly chapters explain what students are learning. This compendium gives the dsDNA Core program a deeper set of labs and real-world exercises to fill full-day work blocks. Each lab ends with a saved artifact, a research log entry, and a short review check.

13.1 Lab 1: Chemical Question Triage

Objective: Teach students to reject questions that cannot be answered with available chemical evidence.

Inputs: Three candidate research interests, the project brief worksheet, and a program-provided example chemical list.

Procedure:

1. For each candidate question, identify the system, comparison, chemical evidence type, expected source tables, and likely limitation.
2. Mark each question as feasible, feasible with narrower scope, or not feasible in 10 weeks.
3. Rewrite weak questions so they specify a chemical set, comparison, evidence type, and output.
4. Choose one question for the internship and one backup question.

Quality gate: The review should make clear what data will be analyzed and what decision the analysis could support.

Real-world exercise: A collaborator asks whether these GC-MS hits can prove that a plant treatment increases medicinal compounds. Students must rewrite the request into a defensible question that separates tentative identity, trait enrichment, and follow-up validation.

13.2 Lab 2: Reproducible Project Build

Objective: Build a project that can be reopened and rerun by another person.

Inputs: RStudio, Git, uafR source or installed package, and `training/scripts/01_project_setup.R`.

Procedure:

1. Run `create_training_project()`.
2. Add a project title, research question, data description, and script order to `README.md`.
3. Create `logs/session_info.txt` using `capture.output(sessionInfo())`.
4. Commit only appropriate files. Confirm that tokens, private data, and local machine paths are absent.

Quality gate: The project runs from its root directory and contains a log explaining what has been done so far.

Real-world exercise: Students exchange project folders and try to identify the correct script order without asking the author. Any confusion becomes a README revision.

13.3 Lab 3: Database Source Audit

Objective: Teach students to interpret database sources as evidence with scope and limitations.

Inputs: Three chemicals from the student's project, PubChem pages or uafR PubChem outputs, KEGG records when available, and the source comparison worksheet.

Procedure:

1. For each chemical, identify a stable name, CID if available, and likely ambiguity.
2. Record which sources provide identity, properties, hazard, occurrence, sensory, biomedical, pathway, reaction, and literature evidence.
3. Mark every source-backed field as analysis-ready, narrative-only, or not relevant.
4. Write one sentence that states what each source can support and one sentence that states what it cannot support.

Quality gate: Students must avoid treating database presence as proof of biological presence in their sample.

Real-world exercise: Students compare caffeine and methyl salicylate. They should notice that one chemical may have strong biomedical and food-context evidence, while another may connect strongly to plant and sensory contexts. The result is a source suitability comparison, not a claim that one chemical is more important.

13.4 Lab 4: Offline Core Workflow

Objective: Teach the GC-MS workflow without depending on live web services.

Inputs: `training/scripts/04_core_workflow.R`, bundled `standard_spread`, and bundled `standard_exacto`.

Procedure:

1. Source the script and run `run_core_workflow(live_lookup = FALSE)`.
2. Inspect the structure of `spread` and `exact`.

3. Identify which columns would be needed to write a methods paragraph.
4. If services are available and live requests are appropriate, rerun with `live_lookup = TRUE` and compare outputs.

Quality gate: Students can explain which objects are bundled teaching outputs and which require live lookup.

Real-world exercise: A student receives a CSV with the wrong column names. They must create an input audit that lists missing columns, suspicious duplicate keys, and a decision on whether the file can be processed.

13.5 Lab 5: Research Enrichment Smoke Test

Objective: Run a small `categorate()` workflow and inspect validation before interpretation.

Inputs: `training/scripts/05_categorate_research.R`, two or three common compounds, and `library_data`.

Procedure:

1. Run a two-compound test with `assay_detail_limit = 0`.
2. Inspect `ValidationSummary`, `TableQuality`, and `SourceCoverage`.
3. Save the result as an RDS file and write the query settings in the research log.
4. Identify three useful tables and three tables that are empty or weak for the chosen compounds.

Quality gate: Students do not inspect `ChemicalTraitMatrix` until they have inspected validation and source coverage.

Real-world exercise: The student must decide whether to run 50 compounds immediately or batch-test 5 first. They should justify the decision using database etiquette, caching, and source coverage.

13.6 Lab 6: Evidence Court

Objective: Train students to defend or withdraw claims based on evidence rows.

Inputs: `ChemicalTraitMatrix`, `ChemicalTraitEvidence`, `ChemicalTraitReport`, and one draft result claim.

Procedure:

1. Choose one matrix column that appears important.
2. Pull all evidence rows for that matrix key.
3. Identify source database, source field, evidence text, source URL, confidence, and extraction rule.
4. Decide whether the trait can support a result, a hypothesis, or only a note.
5. Rewrite the claim using the correct strength of evidence.

Quality gate: Every accepted claim must trace to evidence and every rejected claim must be documented.

Real-world exercise: Students hold a mock review panel. One student argues for a claim, one argues against it, and one acts as editor. The editor decides whether the claim belongs in Results, Discussion, or Future Work.

13.7 Lab 7: Trait Matrix Design

Objective: Build matrices that answer a research question rather than maximize column count.

Inputs: `ChemicalTraits`, `ChemicalTraitOntology`, `chemicalTraitMatrix()`, and the grouping dictionary worksheet.

Procedure:

1. Build a core binary matrix with medium confidence or higher.
2. Build a wider confidence-weighted matrix.
3. Compare matrix dimensions, sparsity, and interpretability.
4. Select a final matrix profile and document why weaker columns were removed.

Quality gate: The final matrix should be smaller, clearer, and more defensible than the widest possible matrix.

Real-world exercise: A project review requests clustering. Students must decide whether the matrix has enough meaningful columns and coverage to make clustering interpretable.

13.8 Lab 8: Figure Clinic

Objective: Convert validated matrices and summaries into figures that support interpretation.

Inputs: `training/scripts/07_visualization.R`, a saved enrichment result, and the figure planning worksheet.

Procedure:

1. Generate a source coverage heatmap and a trait matrix heatmap.
2. Build one additional figure that matches the project question.
3. Write a caption that names the source table, evidence type, and limitation.
4. Peer review the figure using the presentation rubric.

Quality gate: A figure must answer a question, not merely display available columns.

Real-world exercise: Students receive two versions of a figure: one visually attractive but overclaiming, one plain but honest. They revise the attractive figure so it remains readable while becoming scientifically defensible.

13.9 Lab 9: Literature-to-Evidence Bridge

Objective: Connect database-derived patterns to peer-reviewed context.

Inputs: A figure, an evidence row, and three papers related to the system or chemical class.

Procedure:

1. For each paper, extract organism or system, method, chemical evidence, main claim, and limitation.
2. Map each paper to a specific project result.
3. Mark whether the paper supports background, interpretation, mechanism, or follow-up.
4. Write a claim-evidence-limitation paragraph.

Quality gate: Literature should sharpen interpretation, not inflate it.

Real-world exercise: A student finds a paper about the same compound but a different organism and analytical method. They must decide whether it belongs in Introduction, Discussion, or not at all.

13.10 Lab 10: Final Handoff Audit

Objective: Ensure the project can survive after the student leaves.

Inputs: Final project folder, `training/scripts/09_reproducibility_check.R`, final report, figures, and saved result objects.

Procedure:

1. Run `check_training_project()`.
2. Confirm every figure can be regenerated from scripts.
3. Confirm every major result has a source table, evidence row, or citation.
4. Move abandoned scratch files into an archive folder or remove them after program review.
5. Write a continuation note for the next student.

Quality gate: A reviewer should be able to rerun or inspect the work without asking the student to explain where files are.

Real-world exercise: Students trade final project folders and perform a handoff audit. The reviewer writes a one-page note naming what they could reproduce, what was unclear, and what should be fixed first.

14 Supplementary Readings and Real-World Exercises

Readings are assigned to support the work, not to fill time. Students should annotate each reading with three items: what the source teaches, how it changes their project decisions, and what limitation or caution it raises.

14.1 Core Reading List

Week	Reading	Reason for assigning it
1	Wilson et al., Good enough practices in scientific computing [10].	Gives students a practical standard for files, scripts, documentation, and collaboration.
2	R Markdown documentation [1].	Helps students see how code, results, and text can live in one reproducible report.
3	PubChem update paper [4] and PUG-REST update [5].	Explains why PubChem is more than a name lookup and why source/provenance matters.
3	PubChem PUG-REST documentation [6].	Shows the programmatic access model behind many chemical data workflows.
3	KEGG API documentation [3].	Teaches students KEGG REST operations and reinforces the three-calls-per-second academic-use limit.
3	KEGG reference resource paper [2].	Gives biochemical context for compounds, pathways, enzymes, reactions, and annotation.
3	LOTUS natural products occurrence database [7].	Helps students interpret natural product occurrence and taxonomy evidence.
4	Metabolomics Standards Initiative chemical analysis guidance [8].	Establishes why chemical identification rigor and reporting detail matter.
8	Viant et al. metabolomics reporting and best practices [9].	Provides a mature example of how metabolomics evidence is used in high-stakes toxicology contexts.

14.2 Reading Annotation Template

For every assigned reading, students should write:

- **Main idea**: one or two sentences.
- **Project consequence**: what the reading changes about the student's data, code, interpretation, or reporting.
- **Caution**: what the reading says or implies that should prevent overclaiming.
- **Useful phrase**: one technical phrase the student should be able to define.
- **Question for discussion**: one specific question raised by the reading.

14.3 Real-World Exercise Bank

14.3.1 Exercise A: The Unmatched Chemical

Scenario: A student has 20 tentative chemical hits. Four do not return strong source coverage.

Task: Build a missingness table. For each unmatched chemical, decide whether to rename, remove, retain as unmatched, search by synonym, or request follow-up validation.

Expected learning: Missing data are not clutter. They are evidence about the limits of the workflow.

14.3.2 Exercise B: The Hazard Code Trap

Scenario: A chemical has safety annotations and GHS-style terms, but the project is ecological.

Task: Decide which safety terms are relevant to the research question and which belong only in a cautionary appendix.

Expected learning: A hazard code can support screening or caution, but it does not automatically explain biological behavior in a sample.

14.3.3 Exercise C: Pathway Does Not Mean Presence

Scenario: KEGG links a compound to a pathway. A student wants to claim that the pathway is active in their sample.

Task: Rewrite the claim at three evidence levels: source-backed context, hypothesis, and confirmed pathway activity.

Expected learning: Pathway annotation is context, not direct measurement of pathway flux or activity.

14.3.4 Exercise D: Natural Product Occurrence

Scenario: LOTUS-derived evidence suggests a compound occurs in several organisms, but the taxonomic context differs from the student's organism.

Task: Separate direct occurrence evidence, related-organism evidence, and broad natural product evidence.

Expected learning: Occurrence records can guide hypotheses while still requiring taxonomic caution.

14.3.5 Exercise E: Sensory Versus Biological Interpretation

Scenario: FEMA or annotation text suggests flavor or odor relevance.

Task: Decide whether the sensory trait belongs in the main result, a descriptive table, or background motivation.

Expected learning: Sensory descriptors are useful grouping variables only when the research question is sensory, food, aroma, behavior, or exposure related.

14.3.6 Exercise F: Literature Conflict

Scenario: One paper reports a chemical as biologically active, but another paper treats it as a contaminant or background compound.

Task: Build a conflict table with system, dose or abundance, method, evidence strength, and relevance to the student's project.

Expected learning: Conflicting literature is handled by context, not by choosing the more exciting citation.

14.3.7 Exercise G: Matrix Sparsity

Scenario: A full trait matrix has hundreds of columns, but most are present in only one compound.

Task: Compare a full matrix, a core matrix, and a medium-confidence capped matrix. Choose one for the report and justify the choice.

Expected learning: More variables can make analysis weaker when they are sparse, redundant, or poorly supported.

14.3.8 Exercise H: The Reviewer Request

Scenario: A reviewer asks, "How do you know these compounds are meaningfully grouped?"

Task: Prepare a response with matrix construction rules, evidence fields, source coverage, and a limitation statement.

Expected learning: A grouping claim needs source-backed variables, not only visual clustering.

14.3.9 Exercise I: Public Data Ethics

Scenario: A collaborator wants to upload unpublished sample metadata to a public repository before approval.

Task: Write a data handling decision memo: what can be shared, what must be withheld, and what metadata can be generalized.

Expected learning: Reproducibility and confidentiality must be balanced deliberately.

14.3.10 Exercise J: Final Handoff Recovery

Scenario: A student inherits a project folder with no README, ambiguous scripts, and several saved result objects.

Task: Reconstruct the workflow order, identify missing context, and write a recovery plan.

Expected learning: The value of documentation becomes obvious when students experience the cost of missing documentation.

14.4 Discussion Prompts

- When is a database result strong enough for a Results section, and when does it belong in Discussion?
- What is the difference between a chemical being present in PubChem and a chemical being present in a biological sample?
- How should a student report a potentially interesting result with weak source coverage?
- What makes a trait matrix scientifically meaningful rather than merely computationally convenient?
- How should a researcher discuss compounds that are tentatively identified but biologically plausible?

14.5 Recommended Online References

- PubChem PUG-REST documentation: `https://pubchem.ncbi.nlm.nih.gov/docs/pug-rest`
- KEGG API documentation: `https://www.genome.jp/kegg/rest/`
- KEGG API manual: `https://www.genome.jp/kegg/rest/keggapi.html`
- PubChem LOTUS data source page: `https://pubchem.ncbi.nlm.nih.gov/source/25132`
- R Markdown documentation: `https://rmarkdown.rstudio.com/`

15 Project Tracks for Student Interests

Students should be able to bring their own interests to the program. These tracks help adapt the same uafR workflow to different fields while preserving a shared quality standard.

15.1 Track 1: Plant, Agriculture, and Natural Products

Good questions:

- Do volatile profiles differ between treatments, cultivars, tissues, or developmental stages?
- Which detected compounds have natural product occurrence evidence?
- Do compounds cluster by sensory traits, pathway context, or functional groups?

High-value tables:

- `ChemicalOccurrences`, `ChemicalTaxonomy`, and `LOTUSProfile`.
- `ChemicalPathwayRoles`, `KEGGPathways`, and `ChemicalTraits`.

Recommended figures: Occurrence heatmap, trait matrix heatmap, pathway role summary, compound class bar chart.

Caution: Natural product occurrence in another organism supports hypothesis generation, not direct evidence of occurrence in the student's sample.

15.2 Track 2: Environmental Exposure and Toxicology

Good questions:

- Which detected chemicals have hazard, exposure, or regulatory signals requiring follow-up?
- Do sample groups differ by safety-relevant trait profiles?
- Which compounds have properties suggesting persistence, volatility, or hydrophobicity?

High-value tables:

- `ChemicalHazards`, `SafetyProfile`, and `PubChemProperties`.
- `ChemicalMeasurementSummary`, `ChemicalTraits`, and `DerivedGroups`.

Recommended figures: Hazard code count, property scatterplot, source coverage heatmap, trait confidence matrix.

Caution: Hazard annotations are screening evidence. Risk requires dose, exposure, route, organism, and context.

15.3 Track 3: Food, Flavor, Aroma, and Sensory Chemistry

Good questions:

- Which compounds have flavor, odor, or sensory descriptors relevant to a food or plant system?
- Do treatment groups differ in sensory-relevant chemical trait profiles?
- Which compounds should be prioritized for sensory validation or standards?

High-value tables:

- `FEMAProfile`, `ChemicalTerms`, and `ChemicalTraits`.
- `PubChemProperties` and `ChemicalMeasurementSummary`.

Recommended figures: Sensory term heatmap, compound abundance versus sensory evidence, trait count by treatment.

Caution: A flavor annotation does not prove perceptibility. Concentration, threshold, matrix, and mixture effects matter.

15.4 Track 4: Biomedical, Pharmacology, and Public Health

Good questions:

- Which compounds have biomedical terms, pharmacologic actions, or assay evidence?
- Are any detected compounds connected to targets, potencies, or bioactivity categories?
- Which compounds require caution before being described as therapeutic, toxic, or mechanistic?

High-value tables:

- `MeSHProfile`, `PubChemBioactivity`, and `ChemicalTraitEvidence`.
- `ChemicalBioactivities`, `ChemicalTargets`, and `ChemicalPotencies`.

Recommended figures: Bioactivity summary table, target evidence table, biomedical trait matrix, evidence ladder plot.

Caution: Bioassay or pharmacologic annotation does not prove clinical relevance or mechanism in the student's system.

15.5 Track 5: Methods, Data Science, and Tool Evaluation

Good questions:

- How do trait matrix settings change clustering or grouping conclusions?
- Which database sources contribute the most useful evidence for a chemical class?
- How robust are conclusions when low-confidence traits are removed?

High-value tables:

- `ValidationSummary`, `TableQuality`, and `SourceDiagnostics`.
- `SourceCoverage`, `ChemicalTraitMatrix`, and `ChemicalTraitSimilarity`.

Recommended figures: Matrix sensitivity comparison, source coverage summary, trait overlap network, confidence distribution.

Caution: Method evaluation projects still need a scientific question. They should not become demonstrations of every available output.

15.6 Track 6: Chemistry Education and Communication

Good questions:

- Which chemical traits are easiest or hardest for new students to interpret correctly?
- Can a small set of compounds teach uncertainty, source coverage, and chemical grouping?
- What visualizations best communicate chemical evidence to non-specialists?

High-value tables:

- `ChemicalTraitReport` and `ChemicalTraitEvidence`.
- `DataDictionary` and `SourceCoverage`.

Recommended products: Teaching dataset, annotated figure set, evidence interpretation guide, mini-workshop.

Caution: Educational products should preserve scientific accuracy. Simplification should not erase uncertainty.

15.7 Capstone Options

Students can finish with one of several products:

- Research report suitable for a dsDNA Core or lab meeting.
- Poster-ready figure set with captions and methods.
- Manuscript-style results section.
- Reusable analysis pipeline for a larger project.
- Database evidence atlas for a chemical class.
- Teaching module for future interns.

15.8 Track Selection Worksheet

Prompt	**Response**
Chosen track	
Why this track fits the student's interest	
Primary uafR tables	
Most important figure	
Most important limitation	
Likely final product	

A Student Worksheets

A.1 Worksheet 1: Project Brief

Purpose

Use this worksheet to convert a broad interest into a research question that can be tested with chemical data, database enrichment, and reproducible analysis.

Prompt	Student response
Working project title	
Research system	
Primary research question	
Comparison or grouping variable	
Starting chemical list source	
Most relevant evidence types	
Expected final product	
Main risk or limitation	

A.2 Worksheet 2: Chemical Intake Table

Chemical	Source	Confidence	Project relevance	Notes

A.3 Worksheet 3: Database Evidence Map

Chemical	Identity	Safety	Occur.	Pathway	Bioact.	Most useful source

A.4 Worksheet 4: Trait Grouping Dictionary

Variable	Source table	Source field	Allowed values	Why it matters

A.5 Worksheet 5: Figure Planning

Prompt	Student response
Figure question	
Source table or matrix	
Rows or observations	
Columns or variables	
Grouping or color variable	
Expected pattern	
Most important limitation	
Draft caption	

A.6 Worksheet 6: Literature and Evidence Synthesis

Chemical	Finding	uafR evidence	Citation	Interpretation and limitation

A.7 Worksheet 7: Peer Review Form

Review question	Reviewer notes
What is the research question?	
What is the strongest result?	
Where is the evidence strongest?	
Where is the evidence weakest?	
Which figure needs the most improvement?	
Which method detail is missing?	
What should the author revise first?	

A.8 Worksheet 8: Daily Research Log

Prompt	Student response
Date and week	
Today's objective	
Script or file changed	
Output produced	
Evidence inspected	
Problem encountered	
Decision made	
Question for discussion	
Tomorrow's first step	

A.9 Worksheet 9: Evidence Court

Prompt	Student response
Draft claim	
Figure or table supporting claim	
uafR source table	
Evidence row or matrix key	
Source database	
Confidence or limitation	
Literature support	
Rewritten claim	
Reviewer decision	

A.10 Worksheet 10: Final Handoff

Handoff item	Location or status
Project README is current	
Raw data are preserved or access is documented	
Processed data are scripted	
Saved uafR result object is present	
Validation summaries are saved	
Trait matrix and grouping dictionary are saved	
Figures regenerate from scripts	
Final report and presentation are saved	
Session information is saved	
Continuation note is complete	

B Assessment Rubrics

B.1 Weekly Deliverable Rubric

Criterion	Exemplary	Developing	Needs revision
Research focus	Question is specific, feasible, and connected to chemical evidence.	Question is relevant but broad or missing a clear comparison.	Question is vague, untestable, or disconnected from available data.
Reproducibility	Scripts run from the project root and outputs are clearly named.	Most work is reproducible but some paths or manual steps remain.	Work cannot be rerun without guessing or manual intervention.
Evidence traceability	Claims link to tables, fields, figures, or citations.	Some claims have evidence but links are incomplete.	Claims are mostly unsupported or based on broad impressions.
Data quality	Missing values, duplicates, and source coverage are inspected.	Some quality checks are present but incomplete.	Data quality issues are not inspected or documented.
Interpretation	Conclusions are cautious, specific, and aligned with evidence.	Interpretation is plausible but overstates some evidence.	Interpretation makes claims the data cannot support.

B.2 Final Report Rubric

Area	High impact	Acceptable	Revise
Abstract	States question, approach, result, limitation, and significance in clear language.	Covers most elements but lacks precision or caveat.	Reads as a general summary without a concrete result.

Area	High impact	Acceptable	Revise
Methods	Includes package version, settings, data source, filtering, validation, and analysis steps.	Includes core workflow but omits some settings or checks.	Too vague to reproduce.
Results	Figures and tables directly answer the research question.	Results are relevant but not fully connected to the question.	Results are descriptive without a clear purpose.
Figures	Professional, readable, scripted, and captioned with source tables.	Understandable but needs formatting or caption improvements.	Hard to read, manually produced, or not traceable.
Discussion	Explains significance, uncertainty, and next steps.	Provides interpretation but limited uncertainty or future direction.	Overclaims or repeats results without interpretation.
Reproducibility	Project package is clean, runnable, and documented.	Mostly organized with minor missing files or unclear paths.	Files are disorganized or analysis cannot be rerun.

B.3 Presentation Rubric

Criterion	What strong performance looks like
Opening	The audience understands the research problem within the first minute.
Workflow explanation	The student explains data source, uafR processing, enrichment, and analysis without excessive technical clutter.
Main result	The central figure is readable and tied to the research question.
Evidence	The student can name the tables, database sources, or citations supporting the result.
Limitations	Uncertainty is explicit and scientifically mature.
Next steps	The proposed follow-up is realistic and connected to the result.

B.4 Review Prompts

- What is the strongest scientific move the student made this week?
- What is the most important reproducibility issue to fix?
- Which claim needs better evidence?
- Which result is ready for a figure, table, or manuscript sentence?
- What should the student do before collecting or analyzing more data?

B.5 Research Log Rubric

Criterion	Full-credit standard
Daily entries	Each workday has an entry with objective, output, evidence inspected, problem, decision, and next step.
Decision traceability	Important changes in question, data, filtering, or interpretation are dated and justified.
Error documentation	Errors include enough context for another student to understand what failed and how it was fixed.
Evidence discipline	Claims in the log refer to source tables, figures, scripts, or citations.
Program usefulness	A reviewer can read the log and understand current project status within 10 minutes.

B.6 Final Package Rubric

Component	High-quality evidence	Common deduction
README	Explains question, data, scripts, outputs, and rerun order.	Reader cannot tell where to start.
Scripts	Numbered, runnable from project root, and write outputs predictably.	Absolute paths, manual steps, or unclear dependencies.
Data	Raw data are preserved and processed data are derived by scripts.	Edited raw files or undocumented manual cleaning.
uafR outputs	Saved result object, validation summary, source coverage, and trait outputs are present.	Only final figures are saved.
Figures	Regenerated by code and captioned with source table and limitation.	Figures are manually edited or overclaim.
Report	Methods, results, discussion, limitations, and next steps are complete.	Report lacks reproducibility details or uncertainty.
Handoff	Continuation note names what to do next and what not to claim.	Future student would need to reconstruct decisions from scratch.

C Program Implementation Notes

C.1 Program Philosophy

This program treats undergraduate interns as early-career researchers. The expectation is not that students arrive knowing R, chemistry, statistics, or manuscript writing. The expectation is that they learn how to make their work traceable, honest, and useful. The second-pass curriculum assumes a full-day internship: students have enough time each week for reading, guided lab work, independent analysis, writing, peer review, and revision.

C.2 Program Setup Before Week 1

Before Week 1, program setup includes:

- A short orientation to the lab or core facility.
- A list of possible research domains and dsDNA Core contacts.
- A small set of example chemicals for demonstrations.
- Confirmation that R, RStudio, Git, and uafR can be installed.
- A current private student bundle with installer, verifier, package archive, and training PDF.
- Guidance on what data are public, private, unpublished, or restricted.

C.3 Suggested Weekly Pacing

Week	Program focus
1	Research questions are narrowed before projects become too large.
2	Project structure, relative paths, and safe token handling are established.
3	Database interpretation and evidence limits are made explicit.
4	GC-MS inputs, duplicates, and tentative identity claims are inspected.
5	Enrichment outputs, validation summaries, and source coverage are reviewed.
6	Grouping variables and extraction rules are reviewed before analysis expands.
7	Figures are tied to clear questions and explicit missing-data interpretation.
8	Major claims are supported by literature and source evidence.
9	Methods are reproducible and results remain restrained.
10	Communication, handoff quality, and next research steps are finalized.

C.4 Suggested Daily Rhythm

Time block	Use
9:00–9:30	Daily check-in, objective, and one concept.
9:30–11:30	Guided lab or reading discussion.
11:30–12:00	Research log update and troubleshooting list.
1:00–3:00	Independent project application or case exercise.
3:00–4:00	Writing, figure captioning, or evidence audit.
4:00–4:30	dsDNA Core or peer checkpoint and next-day plan.

C.5 Minimum Viable Project

If a student struggles or time is limited, scope is reduced rather than lowering quality. A strong minimum project can use 8 to 15 chemicals, one research question, one trait matrix, two figures, and one interpretation memo. The student still preserves reproducibility, source evidence, and uncertainty.

C.6 Advanced Project Options

Students who move quickly can add:

- Comparison of trait profiles across treatments or sample groups.
- A curated literature table for a chemical class.
- A pathway-focused KEGG interpretation.
- A safety or exposure screening appendix.

- A natural product occurrence analysis using LOTUS-derived fields.
- A reusable R function for project-specific filtering or plotting.

C.7 Database Etiquette

Live database practice is deliberately conservative:

- Start with small smoke tests.
- Use caching for repeated work.
- Use throttling for broad queries.
- Save result objects so students do not repeat identical requests.
- Use offline saved examples when teaching in a room with unstable internet.

C.8 Common Coaching Moments

Issue	Program response
Student overclaims tentative GC-MS hits	Evidence for confirmed identity and tentative status is reviewed before claims move forward.
Matrix has many columns but no interpretation	The student narrows the work through a project-specific grouping dictionary.
Database text is copied into results	The result is rewritten as discrete extracted values with source-backed fields.
Unmatched chemicals disappear	The project adds a missing-data and source-coverage note.
Figures are decorative	The figure is tied to a specific question and research decision.
Script only runs on one laptop	Absolute paths are removed and the workflow is rerun from the project root.

C.9 Recommended Grading Weight

Component	Weight	Notes
Weekly deliverables	30%	Completion, quality, and responsiveness to feedback.
Reproducible project package	20%	File structure, scripts, logs, and rerunnable outputs.
Final report	25%	Scientific logic, methods, results, and interpretation.
Presentation	15%	Clear communication and ability to answer questions.
Research professionalism	10%	Documentation, integrity, collaboration, and growth.

C.10 Final Presentation Questions

These questions are used during final presentations:

1. What was your original question?
2. Which chemical evidence changed your thinking?
3. Which result is strongest?
4. Which result is most uncertain?
5. What should the next student or collaborator do next?

D dsDNA Core Review Notes and Answer Guidance

D.1 Facilitating Without Doing the Project for the Student

In dsDNA Core sessions, evidence stays at the center before answers. The strongest teaching question is often, "Which table, row, script, or citation supports that?" Students learn to diagnose problems by moving from output to source, not by guessing.

D.2 Daily Check-In Structure

This five-minute structure works at the start or end of each day:

1. What did you produce yesterday?
2. What evidence supports the main thing you think you learned?
3. What is uncertain, broken, missing, or confusing?
4. What will you produce by the end of today?
5. What decision do you need from the program lead or dsDNA Core mentor?

D.3 Expected Answers by Program Stage

Stage	Strong student answer	Needs coaching
Question design	Names the system, comparison, chemical evidence, and limitation.	Says only that they want to see what chemicals are important.
Database interpretation	Distinguishes identity, source annotation, occurrence, pathway, hazard, and assay evidence.	Treats any database hit as equivalent support.
GC-MS workflow	Explains match factor, retention time, base peak, area, and tentative identity caveats.	Says the software identified the compound without caveat.
Enrichment	Starts with validation, source coverage, and provenance.	Jumps directly to the most exciting trait or pathway.

Stage	Strong student answer	Needs coaching
Trait matrix	Explains source fields, confidence, matrix keys, and why weak traits were removed.	Uses every available column because more data seem better.
Figure interpretation	Ties each figure to a question and a source table.	Makes decorative plots without a decision or claim.
Literature integration	Uses papers to contextualize results and limitations.	Searches only for papers that support the desired conclusion.
Final report	Methods can be rerun and claims are traceable.	Reads like a narrative without reproducible evidence.

D.4 Common Program Responses

Student pattern	Program response
Overly broad project	The work is limited to one chemical list, one comparison, and one main evidence family.
Too much time cleaning messy data	The student switches to a program-reviewed case dataset for training, then returns to the real data later.
Interpreting before validation	A validation summary screenshot or table is reviewed before biological meaning is discussed.
Matrix is too wide	The matrix is narrowed to medium-confidence traits tied to the research question.
Weak writing	The student writes one claim, one evidence sentence, and one limitation sentence for each figure.
No progress after errors	The workflow moves to a small reproducible example with the exact error, input, expected output, and actual output.

D.5 Review Notes for Real-World Exercises

D.5.1 Unmatched Chemical

A good answer does not delete unmatched chemicals silently. It should preserve them in a missingness table and identify the next action: synonym search, identifier correction, manual review, or follow-up validation.

D.5.2 Hazard Code Trap

A good answer separates hazard screening from risk interpretation. Students should state that risk requires exposure, dose, route, organism, and context.

D.5.3 Pathway Does Not Mean Presence

A good answer rewrites "this pathway is active" as "this compound has KEGG pathway context consistent with..." unless independent pathway activity evidence exists.

D.5.4 Natural Product Occurrence

A good answer separates direct taxonomic evidence from broad natural product context. Students should not imply occurrence in their organism unless the source supports it.

D.5.5 Matrix Sparsity

A good answer favors a smaller matrix if it has clearer interpretation, better confidence, and sufficient coverage across compounds. Students should explain what was removed and why.

D.6 When To Slow the Program Down

The program slows down when:

- Students cannot explain their research question without reading notes.
- Source coverage is weak and students are ignoring it.
- Scripts contain absolute paths or manual steps that cannot be repeated.
- Figures are being made before variables are defined.
- Literature is being used as decoration rather than evidence.

D.7 When To Accelerate

The program accelerates when:

- The student can rerun their workflow from a clean R session.
- The grouping dictionary is reviewed and matrix outputs are stable.
- Figures have clear captions and limitations.
- The student can explain what would change their interpretation.

D.8 Final Review Questions

Before the final package is accepted, the review asks:

1. Which result is strongest, and what evidence supports it?
2. Which result is most uncertain, and why?

3. Which chemicals were excluded, unmatched, or weakly supported?
4. Which database source was most useful, and which was least useful?
5. What should the next student do first?
6. What claim should not be made from this project?

E Installation and Setup Appendix

E.1 Recommended Student Installation

The recommended student installation path is a private uafR student bundle distributed by the DSU dsDNA Core program. This keeps the package repository private and gives every student the same RStudio-based setup.

The student bundle includes:

- `START_HERE.md`, the first checklist to open.
- A local uafR source package archive in `packages/`.
- `preflight_check.R`, the machine readiness check.
- `install_uafR_from_bundle.R`, the student installer.
- `update_uafR_from_bundle.R`, the reinstall/update script for newer bundles.
- `verify_uafR_install.R`, the post-install verification script.
- `run_student_acceptance_test.R`, the offline readiness test.
- The current training manual PDF.
- Training scripts and data templates used during the internship.

The installer still needs internet access for CRAN and Bioconductor dependencies. The uafR package itself is installed from the local bundle archive.

E.2 Install From the Student Bundle

The installation workflow is:

1. Install current R and RStudio.
2. Unzip the whole bundle folder.
3. Open RStudio.
4. Open `START_HERE.md` from the unzipped bundle folder.
5. Source `preflight_check.R`.
6. If preflight has no failed checks, source `install_uafR_from_bundle.R`.

7. Source `run_student_acceptance_test.R`.
8. Open `training/uafR_training_manual.pdf`.

The three main commands are:

```
source("preflight_check.R")
source("install_uafR_from_bundle.R")
source("run_student_acceptance_test.R")
```

The installer prints five visible steps: checking the bundle and R library, installing CRAN dependencies, installing Bioconductor dependencies, installing uafR from the local archive, and verifying the installation. The installer writes `uafR_install_log.txt` in the bundle folder.

E.3 Create a New Student Bundle

From the repository root, the program build step is:

```
Rscript tools/build_student_bundle.R
```

This creates a versioned bundle folder and zip archive under `student_bundle/`. The zip archive can be copied to a USB drive, shared through an approved campus file-transfer method, or unpacked onto classroom machines.

E.4 Confirm the Installation Without Live Web Queries

The verification script checks package loading, required dependencies, bundled datasets, and the saved standard categorate result. It does not query live web services.

```
source("verify_uafR_install.R")
```

E.5 Student Acceptance Test

The acceptance test confirms that the installed package is ready for the training workflow. It runs package verification, loads the core data objects, and runs the offline workflow from `training/scripts/04_core_workflow.R`. It should end with:

```
Acceptance test result: PASS.
```

```
source("run_student_acceptance_test.R")
```

E.6 Optional Live Smoke Test

After the offline verification passes, run a two-compound live test when internet access is available:

```
library(uafR)
data("library_data", package = "uafR")

quick_result <- categorate(
  compounds = c("aspirin", "caffeine"),
  chemical_library = library_data,
  input_format = "wide",
  detail = "research",
  cache = FALSE,
  throttle = 0.1,
  assay_detail_limit = 0
)

validateCategorateResult(quick_result)$Summary
quick_result$ChemicalTraitReport
```

The acceptance test can also include the live smoke test:

```
Sys.setenv(UAFR_STUDENT_LIVE = "1")
source("run_student_acceptance_test.R")
```

E.7 Troubleshooting

Problem	Likely cause	What to do
Installer cannot find uafR archive	The bundle was not fully unzipped or the `packages/` folder was moved.	Unzip the whole bundle again and keep the folder structure unchanged.
Preflight says the R library is not writable	R cannot install packages into the first library path.	Restart RStudio. If the problem remains, set a user library before installing.
Bioconductor dependency fails	Internet access, R version, or Bioconductor access is blocked.	Run `preflight_check.R`, update R if it is old, and try again on a network that can reach Bioconductor.
R asks about updating many packages	RStudio or Bioconductor is offering broad updates.	Choose the option that does not update unrelated packages.
`library(uafR)` fails right after install	RStudio has not refreshed the library.	Restart RStudio, then run `library(uafR)`.
No library detected	The chemical library argument was not supplied to `categorate()`.	Load `library_data` and pass `chemical_library = library_data`.
Long database query	Live PubChem, KEGG, LOTUS, or other services are responding slowly.	Start with one or two compounds, keep `assay_detail_limit = 0`, use caching, and scale up after the smoke test works.
Missing output tables	A live lookup failed or returned incomplete source data.	Run `validateCategorateResult()` and inspect `ValidationSummary`, `TableQuality`, and `SourceDiagnostics`.

F uafR Table and Script Guide

F.1 High-Value categorate Tables

Table	Research use
`PubChemIdentity`	Check CID matches, match status, and source URLs.
`PubChemProperties`	Compare formula, exact mass, molecular weight, and descriptor fields when available.
`SourceCoverage`	Identify which databases contributed evidence for each chemical.
`ChemicalTerms`	Inspect long-form extracted terms before committing to variables.
`ChemicalTraits`	Use cross-source discrete traits with confidence and evidence fields.
`ChemicalTraitOntology`	Use whitelisted ontology-style traits grouped by research domain.
`ChemicalTraitMatrix`	Analyze chemicals by binary, count, or confidence-weighted trait columns.
`ChemicalTraitEvidence`	Trace matrix columns back to source evidence.
`ChemicalTraitReport`	Review one compact row per compound for high-level interpretation.
`ChemicalMeasurementSummary`	Summarize extracted measured or computed properties.
`ChemicalHazards`	Inspect source-backed safety and hazard values.
`ChemicalOccurrences`	Review organism or natural product occurrence evidence.
`ChemicalPathwayRoles`	Interpret KEGG pathway context and biochemical roles.
`KEGGReactionParticipants`	Examine reaction participation and related biochemical context.
`DataDictionary`	Confirm expected table schemas and field meaning.
`TableQuality`	Audit missing columns, duplicate keys, and completeness.
`ValidationSummary`	Start every review here before interpreting output.

F.2 Training Scripts

Script	Purpose
`00_install_check.R`	Confirms R, package, dependencies, and example data.
`01_project_setup.R`	Creates a reproducible student project folder.
`04_core_workflow.R`	Runs the core GC-MS example workflow.
`05_categorate_research.R`	Runs a small research enrichment smoke test.
`06_trait_matrix_analysis.R`	Builds selected trait matrices and evidence tables.
`07_visualization.R`	Creates example coverage and trait figures.
`09_reproducibility_check.R`	Audits a student project folder before final submission.
`run_training_smoke_tests.R`	Runs lightweight checks across the training scripts.

F.3 Student Bundle Scripts

Script	Purpose
`tools/build_student_bundle.R`	Builds the private student bundle folder and zip archive from the repository root.
`START_HERE.md`	Gives the first student checklist for setup, installation, acceptance testing, and opening the manual.
`preflight_check.R`	Checks R, RStudio, internet access, bundle contents, library write access, and required dependencies before installation.
`install_uafR_from_bundle.R`	Installs dependencies and the local uafR source archive from the unzipped student bundle.
`update_uafR_from_bundle.R`	Reinstalls uafR from a newer bundle and verifies the result.
`verify_uafR_install.R`	Confirms package loading, dependencies, bundled datasets, and saved standard categorate data without live web queries.
`run_student_acceptance_test.R`	Runs package verification, the offline training workflow, and an optional live database smoke test.
`examples/test_install.R`	Reruns the acceptance test from the bundle examples folder.

F.4 Choosing Detail Levels

- Use `detail = "standard"` for the legacy eight-table output.
- Use `detail = "research"` for most student projects. It gives rich tables while keeping the query smaller than full assay detail.

- Use `detail = "full"` only when the project needs broad assay details or deeper PubChem annotations.

F.5 Recommended Smoke Test Settings

```
categorate(
  compounds = c("aspirin", "caffeine"),
  chemical_library = library_data,
  input_format = "wide",
  detail = "research",
  cache = TRUE,
  throttle = 0.2,
  assay_detail_limit = 0,
  trait_matrix_profile = "core",
  trait_matrix_min_confidence = "medium",
  trait_matrix_max_traits = 100
)
```

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